Supplementary Material

Confidence Calibration and Uncertainty-Aware Prediction for Reliable Skin Lesion Classification

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This document extends the main paper with details that did not fit within its page limit. All numbers are generated from saved per-image predictions by the scripts in the code repository. E0/E1: ResNet-50 trained with plain or class-weighted cross-entropy; seed 42 is the model of the submitted manuscript. Source collections: HAM10000 [1], BCN_20000 [2] and MSK [3].

S1 Data, Sources and Duplicate Audit

Table S1: Images per source collection and per class in the lesion-grouped split.

	Train	Val.	Test
HAM10000	7964	986	1065
BCN_20000	9935	1302	1176
MSK	2336	285	282
MEL	3579	448	495
NV	10325	1300	1250
BCC	2638	349	336
AK	683	93	91
BKL	2100	270	254
DF	197	22	20
VASC	208	24	21
SCC	505	67	56
Total	20235	2573	2523

Lesion identifiers are available for 23,247 of the 25,331 images. They cover all HAM10000 and BCN_20000 images and 819 MSK images; the remaining 2,084 MSK images have no identifier and were treated as single-image groups. The split uses 13,931 groups, 5,059 of which contain more than one image; no group occurs in two partitions.

Duplicate screen. Exact duplicates were found by MD5 hashes of the files and of the decoded 224×224 pixels: 50 groups of pixel-identical images, 4 of which cross partitions. Near-duplicates were found with a 64-bit DCT perceptual hash and the cosine similarity of ImageNet ResNet-50 embeddings. Candidate pairs had a hash distance ≤ 6 or a cosine ≥ 0.95, and a pair was flagged when the hash distance was ≤ 4 and the cosine ≥ 0.97. Within partitions, 97% of the 428 flagged pairs share a lesion identifier, which validates the rule. Across partitions it flags 10 pairs (7 near pixel-identical) involving 8 internal-test images. Visual inspection showed these to be re-uploads of the same image under different identifiers. Removing them gives balanced accuracy 0.565 (all images: 0.5653). Removing every test image with any cross-partition candidate of cosine ≥ 0.97 (276 images) gives 0.576. Between the internal data and the official test set there are no exact duplicates and 2 flagged near-duplicates; excluding them changes the official known-class balanced accuracy from 0.491 to 0.491.

Table S2: Internal-test performance of ResNet-50 E1 by source collection (temperature and τ from validation). Balanced accuracy is averaged over classes with $\geq$5 test images in that source.

Source	n	Acc.	Bal. acc.	ECE$_T$	Cov.	Sel. acc.	Classes
HAM10000	1065	0.819	0.652	0.093	0.904	0.869	MEL, NV, BCC, AK, BKL, DF, VASC, SCC
BCN_20000	1176	0.645	0.456	0.067	0.857	0.688	MEL, NV, BCC, AK, BKL, DF, VASC, SCC
MSK	282	0.691	0.705	0.065	0.922	0.712	MEL, NV, BKL

S2 Training Configuration

Table S3: Training configuration (identical for all runs except the seed).

Input	224×224 RGB, bicubic resize, ImageNet mean/std
Augmentation (train)	horizontal flip, rotation $\pm 15^{\circ}$, scale $\pm 10\%$, translation $\pm 5\%$, brightness/contrast ± 0.1
Optimiser	AdamW, weight decay 10^{-4}, batch 32, gradient clipping 1.0, no label smoothing
Phase 1	backbone frozen, 10 epochs, learning rate 10^{-3}
Phase 2	last 30% of backbone parameters unfrozen, up to 15 epochs, learning rate 5×10^{-5}
Schedule / stopping	ReduceLROnPlateau on validation macro-F1 (factor 0.5, patience 3); early stopping on validation macro-F1 (patience 6)
Class weights (E1)	$w_c = N/(Kn_c)$ from the training split: NV 0.24, MEL 0.71, BCC 0.96, BKL 1.20, AK 3.70, SCC 5.01, VASC 12.2, DF 12.8
Hardware / software	NVIDIA RTX 4050 Laptop GPU (6 GB), PyTorch 2.11, CUDA 12.8; about 1 h per run

Seed 42 corresponds to the models of the submitted manuscript; seeds 43–45 were retrained with an identical protocol. For speed, training read images from a lossless cache of the 224×224 resize. This is bit-identical to resizing on the fly, as verified on 200 images.

S3 Checkpoint Rule and Seed-Level Statistics

The stored weights of the submitted models are those of the epoch at which early stopping triggered: epoch 24 for E1 (best validation macro-F1 at epoch 18) and epoch 21 for E0 (best at epoch 15). EfficientNet-B3 ran all 25 epochs, and its last epoch was also its best. For the new seeds both checkpoints were stored. The two rules differ only when early stopping triggers before the last epoch, which happened in 1 of 3 new E1 runs. Averaged over those runs, the best-epoch weights had a lower temperature (2.170 vs. 2.467) and a lower uncalibrated ECE (0.101 vs. 0.124). Balanced accuracy (0.568 vs. 0.574), AURC (0.131 vs. 0.132) and calibrated NLL (0.815 vs. 0.814) were similar.

Table S4: Seed-42 models (as submitted) on the internal test split with 95% lesion-cluster bootstrap intervals (2,000 resamples of 1,399 lesion groups). The last column is the paired difference.

Metric	E0 (plain CE)	E1 (weighted CE)	E1 − E0
Accuracy	0.732 [0.704, 0.759]	0.724 [0.698, 0.748]	-0.009 [-0.027, +0.010]
Balanced accuracy	0.537 [0.485, 0.598]	0.565 [0.513, 0.623]	+0.029 [-0.005, +0.065]
Macro-F1	0.553 [0.501, 0.600]	0.557 [0.504, 0.600]	+0.004 [-0.029, +0.039]
ROC-AUC (macro, OvR)	0.924 [0.905, 0.941]	0.919 [0.902, 0.934]	-0.006 [-0.018, +0.009]
ECE after T	0.031 [0.025, 0.054]	0.050 [0.039, 0.073]	+0.019 [-0.006, +0.037]
NLL after T	0.753 [0.690, 0.820]	0.811 [0.744, 0.882]	+0.058 [+0.018, +0.098]
Error AUROC (MSP)	0.785 [0.759, 0.811]	0.761 [0.732, 0.789]	-0.024 [-0.051, +0.002]
AURC (MSP)	0.110 [0.092, 0.131]	0.134 [0.109, 0.160]	+0.024 [+0.007, +0.040]
Coverage (85% val. target)	0.868 [0.850, 0.886]	0.884 [0.868, 0.900]	+0.016 [-0.000, +0.033]
Accepted-set accuracy	0.780 [0.754, 0.805]	0.769 [0.743, 0.793]	-0.012 [-0.032, +0.008]

Table S5: Mean ± SD over independently trained seeds (seed 42 = submitted run; further seeds retrained with the identical protocol). All post-hoc maps and thresholds fitted on validation. Last column: mean per-seed difference, with (number of seeds where E1 > E0 / seeds paired, paired t-test p). †Official ISIC 2019 test set, never used during development (8 known classes; UNK = out-of-distribution).

Metric (internal test unless marked)	E0 ResNet-50 (n=3)	E1 ResNet-50 (n=4)	E1 EffNet-B3 (n=3)	E1−E0 paired
Accuracy	0.727 ± 0.006	0.723 ± 0.006	0.684 ± 0.005	-0.006 (1/3, p=0.28)
Balanced accuracy	0.533 ± 0.004	0.572 ± 0.014	0.543 ± 0.008	+0.041 (3/3, p=0.07)
Macro-F1	0.551 ± 0.004	0.560 ± 0.004	0.502 ± 0.013	+0.010 (3/3, p=0.09)
ROC-AUC	0.924 ± 0.001	0.921 ± 0.005	0.920 ± 0.001	-0.004 (1/3, p=0.20)
ECE (uncalibrated)	0.190 ± 0.011	0.130 ± 0.018	0.083 ± 0.004	-0.053 (0/3, p=0.05)
ECE (T)	0.027 ± 0.006	0.052 ± 0.010	0.043 ± 0.011	+0.020 (3/3, p=0.02)
NLL (T)	0.765 ± 0.011	0.814 ± 0.005	0.870 ± 0.016	+0.047 (3/3, p=0.01)
Brier (T)	0.380 ± 0.005	0.399 ± 0.003	0.438 ± 0.006	+0.018 (3/3, p=0.00)
Temperature T	3.263 ± 0.332	2.578 ± 0.245	1.660 ± 0.100	-0.628 (0/3, p=0.17)
Error AUROC, MSP	0.788 ± 0.004	0.760 ± 0.007	0.761 ± 0.012	-0.029 (0/3, p=0.05)
Error AUROC, MC entropy	0.773 ± 0.004	0.750 ± 0.008	0.761 ± 0.011	-0.026 (0/3, p=0.03)
AURC, MSP	0.115 ± 0.005	0.133 ± 0.006	0.152 ± 0.006	+0.020 (3/3, p=0.01)
Coverage at val. 85%	0.869 ± 0.009	0.873 ± 0.014	0.869 ± 0.004	+0.003 (2/3, p=0.83)
Selective accuracy	0.777 ± 0.004	0.768 ± 0.006	0.731 ± 0.004	-0.012 (0/3, p=0.10)
Official test bal. acc.†	0.469 ± 0.013	0.506 ± 0.015	0.523 ± 0.002	+0.040 (3/3, p=0.10)
Official test ECE (T)†	0.018 ± 0.004	0.032 ± 0.010	0.019 ± 0.006	+0.010 (2/3, p=0.31)
UNK detection AUROC (MSP)†	0.660 ± 0.020	0.623 ± 0.015	0.618 ± 0.004	-0.042 (0/3, p=0.03)

Table S6: Which model is best depends on the reliability dimension (means over seeds; internal test unless stated).

Dimension	R50-E0	R50-E1	B3-E1	Best
Balanced accuracy	0.533	0.572	0.543	R50-E1
Accuracy	0.727	0.723	0.684	R50-E0
Macro-F1	0.551	0.560	0.502	R50-E1
ROC-AUC	0.924	0.921	0.920	R50-E0
ECE, uncalibrated	0.190	0.130	0.083	B3-E1
ECE after T	0.027	0.052	0.043	R50-E0
NLL after T	0.765	0.814	0.870	R50-E0
Error AUROC (MSP)	0.788	0.760	0.761	R50-E0
AURC (MSP)	0.115	0.133	0.152	R50-E0
Official test bal. acc.	0.469	0.506	0.523	B3-E1
Official test ECE_T	0.018	0.032	0.019	R50-E0
UNK detection AUROC	0.660	0.623	0.618	R50-E0

S4 Calibration Details

ECE uses 15 equal-width bins; adaptive ECE (aECE) uses 15 equal-mass bins and class-wise ECE (cwECE) averages the ECE of each class probability [4, 5]. Prior correction subtracts $\log w_c$ from the logits of E1 (and logit adjustment adds it to E0) following [6]. Calibration-aware training losses such as focal loss [7] were not evaluated.

Table S7: Calibration on the internal test split (seed 42). All maps are fitted on validation only. ECE: 15 equal-width bins; aECE: 15 equal-mass bins; cwECE: class-wise ECE. Vector scaling and prior correction change the decision rule, hence the last column.

Model	Post-hoc map	ECE	aECE	cwECE	Brier	NLL	Bal. acc.
ResNet-50 E1	none	0.149	0.148	0.042	0.425	1.264	0.565
	temperature	0.050	0.047	0.032	0.394	0.811	0.565
	vector scaling	0.038	0.041	0.015	0.375	0.768	0.540
	prior correction	0.139	0.139	0.038	0.409	1.305	0.533
	prior corr. + T	0.065	0.066	0.031	0.384	0.803	0.533
ResNet-50 E0	none	0.179	0.177	0.046	0.438	1.229	0.537
	temperature	0.031	0.029	0.014	0.375	0.753	0.537
	vector scaling	0.025	0.023	0.011	0.375	0.751	0.505
EffNet-B3 E1	none	0.086	0.087	0.041	0.452	0.990	0.536
	temperature	0.055	0.057	0.038	0.443	0.882	0.536
	vector scaling	0.024	0.024	0.014	0.392	0.792	0.444
	prior correction	0.092	0.089	0.026	0.409	0.936	0.457
	prior corr. + T	0.046	0.048	0.026	0.401	0.819	0.457

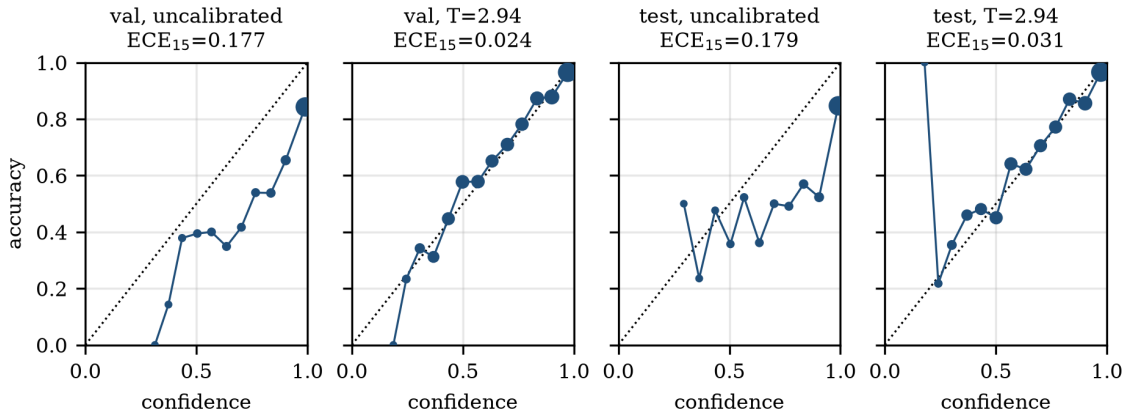


Figure S1: Reliability diagrams of ResNet-50 E0 (seed 42) on validation and internal test, before and after temperature scaling.

S5 Selective Prediction Details

Table S8: Validation-locked selective prediction (ResNet-50 E1, seed 42). Thresholds are chosen on validation for a target validation coverage and applied unchanged to test. Rej. acc.: accuracy among rejected test cases; MEL sens.: melanoma sensitivity among accepted cases (all cases: 0.693).

Score	Target	τ	Val. cov.	Test cov.	Sel. acc.	Rej. acc.	MEL sens.	MEL rej.
MSP (cal.)	0.70	0.524	0.690	0.723	0.812	0.493	0.756	147
MSP (cal.)	0.80	0.463	0.802	0.841	0.786	0.397	0.742	72
MSP (cal.)	0.85	0.429	0.851	0.884	0.769	0.380	0.733	53
MSP (cal.)	0.90	0.391	0.900	0.922	0.752	0.383	0.711	32
MSP (cal.)	0.95	0.348	0.941	0.950	0.744	0.333	0.706	18
MSP (raw)	0.70	0.798	0.691	0.734	0.809	0.487	0.790	161
MSP (raw)	0.80	0.674	0.801	0.836	0.781	0.434	0.755	95
MSP (raw)	0.85	0.616	0.851	0.878	0.764	0.432	0.732	66
MSP (raw)	0.90	0.564	0.901	0.912	0.756	0.395	0.728	47
MSP (raw)	0.95	0.518	0.941	0.947	0.745	0.338	0.709	18
MC entropy	0.70	1.187	0.700	0.746	0.798	0.506	0.741	113
MC entropy	0.80	1.330	0.800	0.828	0.773	0.488	0.726	72
MC entropy	0.85	1.419	0.850	0.874	0.763	0.455	0.719	53
MC entropy	0.90	1.517	0.900	0.922	0.749	0.423	0.704	32
MC entropy	0.95	1.622	0.950	0.956	0.737	0.432	0.697	16

Table S9: Class-wise behaviour of the validation-locked rule ($\tau = 0.429$) on the internal test split.

Class	n	Coverage	Rejected	Recall (all)	Recall (accepted)
MEL	495	0.893	53	0.693	0.733
NV	1250	0.929	89	0.814	0.842
BCC	336	0.821	60	0.744	0.797
AK	91	0.758	22	0.220	0.232
BKL	254	0.827	44	0.626	0.681
DF	20	0.800	4	0.450	0.562
VASC	21	0.905	2	0.762	0.737
SCC	56	0.679	18	0.214	0.289

Table S10: Risk-controlled thresholds (SGR, Geifman & El-Yaniv) fitted on validation with $\delta = 0.05$ and applied to test.

Model	r^*	θ	Bound	Val. cov.	Test cov.	Test risk
ResNet-50 E1	0.10	0.770	0.099	0.342	0.386	0.086
ResNet-50 E1	0.15	0.643	0.150	0.503	0.555	0.136
ResNet-50 E1	0.20	0.532	0.200	0.674	0.710	0.181
ResNet-50 E1	0.25	0.457	0.250	0.813	0.848	0.218
ResNet-50 E0	0.10	0.806	0.099	0.421	0.434	0.079
ResNet-50 E0	0.15	0.667	0.150	0.588	0.607	0.130
ResNet-50 E0	0.20	0.548	0.200	0.731	0.753	0.172
ResNet-50 E0	0.25	0.417	0.250	0.886	0.897	0.229
EffNet-B3 E1	0.10	0.791	0.100	0.299	0.327	0.117
EffNet-B3 E1	0.15	0.651	0.150	0.457	0.487	0.151
EffNet-B3 E1	0.20	0.576	0.200	0.560	0.584	0.179
EffNet-B3 E1	0.25	0.496	0.250	0.689	0.711	0.212

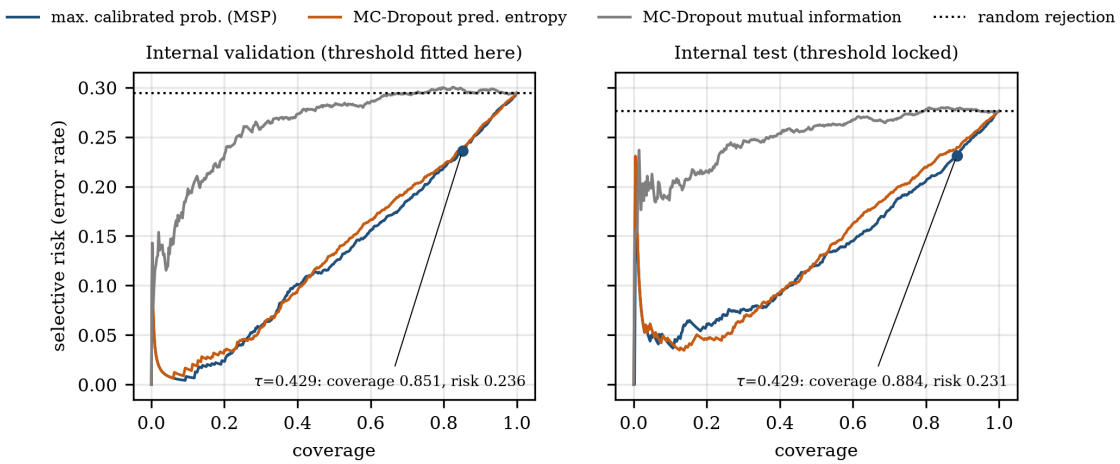


Figure S2: Risk–coverage curves of ResNet-50 E1 (seed 42) on validation, where the threshold is fitted, and on the internal test split, where it is locked.

S6 Uncertainty Scores, Conformal Sets and Official Test Set

Mutual information and expected entropy follow the standard decomposition of predictive entropy [8]. Conformal sets use the LAC score [9], the adaptive APS score [10] and a class-conditional (Mondrian) variant of LAC. Selective risk is summarised by AURC and E-AURC [11], and risk-controlled thresholds use SGR [12].

Table S11: Error detection on the internal test split (ResNet-50 E1, seed 42). Target: errors of the deployed (deterministic) classifier. Δ: paired lesion-cluster bootstrap difference to calibrated MSP. pAUROC: standardised partial AUROC for FPR ≤ 0.2. AURC: area under the risk–coverage curve (random rejection = 0.276).

Score	AUROC	Δ vs MSP [95% CI]	pAUROC	AURC
MSP ($1 - \max_k p_k$), calibrated	0.761	–	0.627	0.134
MSP, uncalibrated	0.745	-0.016 [-0.025, -0.007]	0.621	0.144
Softmax entropy, calibrated	0.750	-0.010 [-0.022, +0.001]	0.601	0.134
MC predictive entropy (paper)	0.745	-0.016 [-0.029, -0.003]	0.598	0.136
MC expected entropy (aleatoric)	0.747	-0.014 [-0.027, -0.002]	0.600	0.136
MC mutual information (epistemic)	0.528	-0.232 [-0.276, -0.189]	0.497	0.249
MC predictive variance	0.620	-0.141 [-0.178, -0.107]	0.534	0.197
MC mean-probability MSP	0.761	+0.000 [-0.003, +0.003]	0.625	0.134
TTA predictive entropy	0.760	-0.000 [-0.016, +0.016]	0.615	0.130
TTA mutual information	0.705	-0.056 [-0.085, -0.028]	0.580	0.154

Table S12: Split-conformal prediction sets (ResNet-50 E1, seed 42; calibration set = validation, evaluation = internal test).

Method	$1-\alpha$	Coverage	Set size	Singleton	Acc. singl.	MEL cov.	Min. class cov.
LAC	0.90	0.915	1.919	0.344	0.897	0.956	0.321
APS	0.90	0.979	3.505	0.040	0.951	0.992	0.732
Class-cond. LAC	0.90	0.902	3.282	0.138	0.916	0.933	0.846
LAC	0.95	0.946	2.562	0.167	0.905	0.976	0.500
APS	0.95	0.987	4.065	0.015	0.895	0.996	0.804
Class-cond. LAC	0.95	0.955	3.774	0.108	0.938	0.962	0.929

Table S13: Untouched evaluation on the official ISIC 2019 test set (8,238 images; 2,047 UNK). Temperature, threshold (τ for 85% validation coverage) and conformal quantile come from the internal validation split. Acc./Bal. acc./ECE on the 6,191 known-class images; OOD = UNK vs. known AUROC; Cov./Sel. acc. on known classes; UNK acc.: fraction of UNK images accepted; BA_9: 9-class balanced accuracy without abstention and with rejected cases labelled UNK; Conf.: LAC coverage at 90% nominal.

Model	Acc.	Bal. acc.	ECE	ECE_T	OOD MSP	OOD MC-H	OOD MI	Cov.	Sel. acc.	UNK acc.	BA_9	BA_9 rej.	Conf.
ResNet-50 E0	0.674	0.478	0.211	0.022	0.683	0.714	0.692	0.812	0.743	0.606	0.425	0.393	0.874
ResNet-50 E1	0.659	0.491	0.197	0.019	0.628	0.684	0.669	0.832	0.718	0.675	0.437	0.426	0.866
EffNet-B3 E1	0.643	0.524	0.114	0.025	0.615	0.681	0.728	0.849	0.693	0.739	0.465	0.446	0.880

S7 Grad-CAM Checks

The submitted version showed four curated Grad-CAM examples [13]. We applied two quantitative checks to 300 random test images, at the original layer (`conv3` of the last bottleneck) and at the output of `layer4`.

- *Cascading parameter randomisation* [14]. Re-initialising only the head already drops the rank correlation between original and randomised maps to 0.04, so the maps depend on the trained weights (Table S14).
- *Deletion/insertion* [15]. Grad-CAM clearly beats a random map. For deletion, however, it is no better than a fixed centre-Gaussian prior that ignores the image (AUC 0.538 vs. 0.538, Wilcoxon $p = 0.86$); for insertion it is only modestly better (0.754 vs. 0.723, $p = 0.0022$) (Table S15).

On average, 55.5% of the map mass falls in the central quarter of the image. Maps of uncertain predictions are more concentrated (Spearman $\rho = -0.58$ between map entropy and MC entropy) and often lie off the lesion (Fig. S3). As an error detector, map concentration reaches an AUROC of only 0.627. Grad-CAM is therefore not used as reliability evidence in the main paper.

Table S14: Cascading model-parameter randomisation (300 random test images): similarity between Grad-CAM maps of the trained and the progressively randomised network. Values near 0 indicate model-sensitive maps.

Randomised	Spearman ρ	Top-10% IoU
up to head	0.043	0.166
up to layer4	0.115	0.078
up to layer3	-0.034	0.048
up to layer2	0.022	0.053
up to layer1	-0.094	0.041

Table S15: Deletion/insertion faithfulness (300 random test images; probability of the predicted class as salient pixels are blurred out / restored in 5% steps). Lower deletion and higher insertion AUC are better.

Saliency	Deletion AUC	Insertion AUC
Grad-CAM (paper layer)	0.538	0.754
Grad-CAM (layer4 output)	0.533	0.760
Dataset-mean Grad-CAM	0.540	0.731
Centre Gaussian prior	0.538	0.723
Random map	0.586	0.579

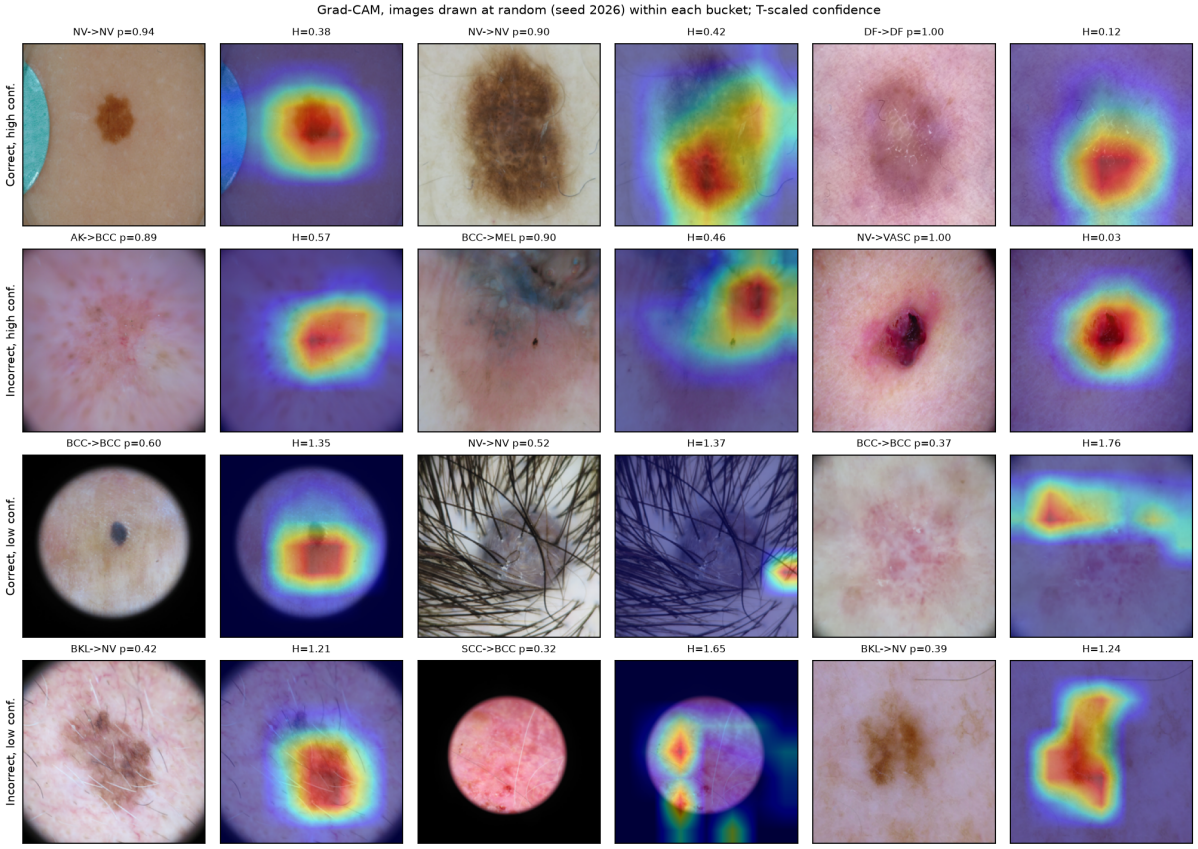


Figure S3: Grad-CAM maps for test images drawn at random (fixed seed) from each correctness/confidence bucket of ResNet-50 E1. Titles: true→predicted class, calibrated confidence and MC entropy H.

S8 Higher-Resolution Replication

[ResNet-50 E0/E1 at 320 × 320 px: training in progress; this section is generated automatically when the runs finish.]

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